RESEARCH SEMINAR

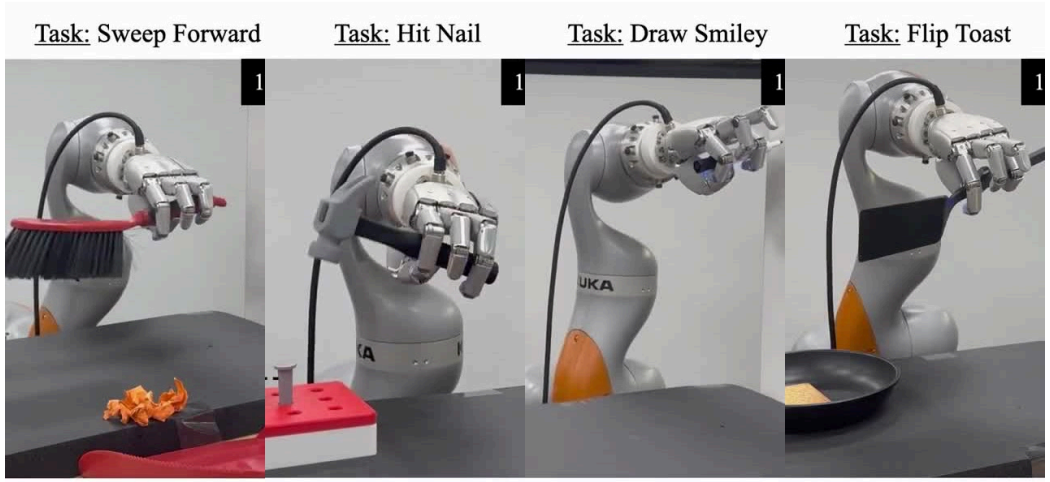
SimToolReal

An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation

Can one reusable controller turn tool motion into robot skill?

Kushal Kedia* · Tyler Ga Wei Lum*
Jeannette Bohg · C. Karen Liu
Cornell University · Stanford University

RSS 2026 · Based on arXiv v2, 24 February 2026



01 / BRAINSTORM

How would you teach a robot to use tools?

Suppose the robot must use
a new brush, hammer or spatula.
What would you build first?

Imitate robot actions

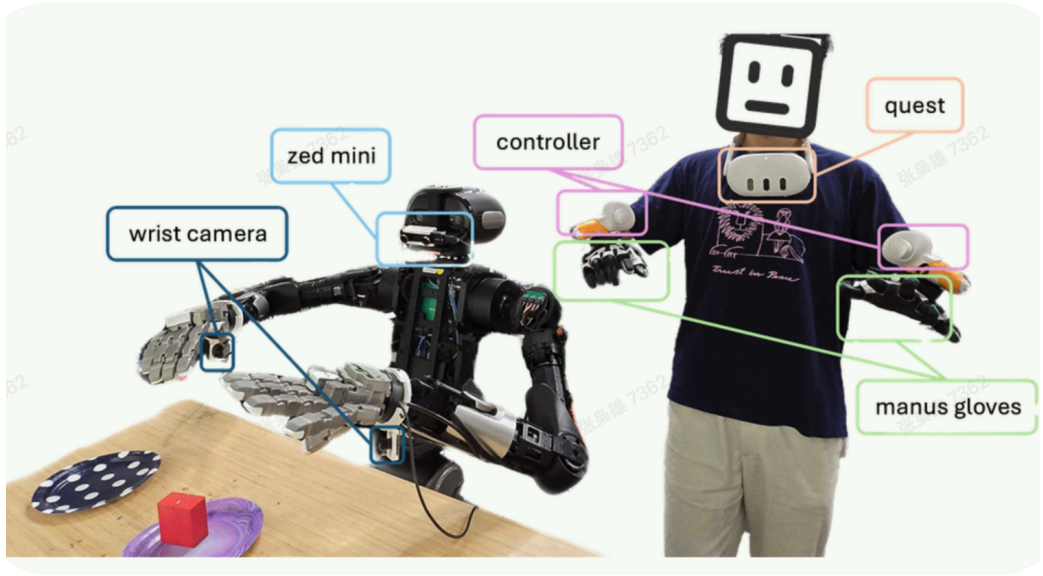
Retarget human motion

Learn through simulation

01 / MOTIVATION

What makes demonstration-based tool use difficult?

Robot demonstrations



Hardware-heavy capture and calibration

Human-to-robot retargeting



Human · stable wraparound grasp



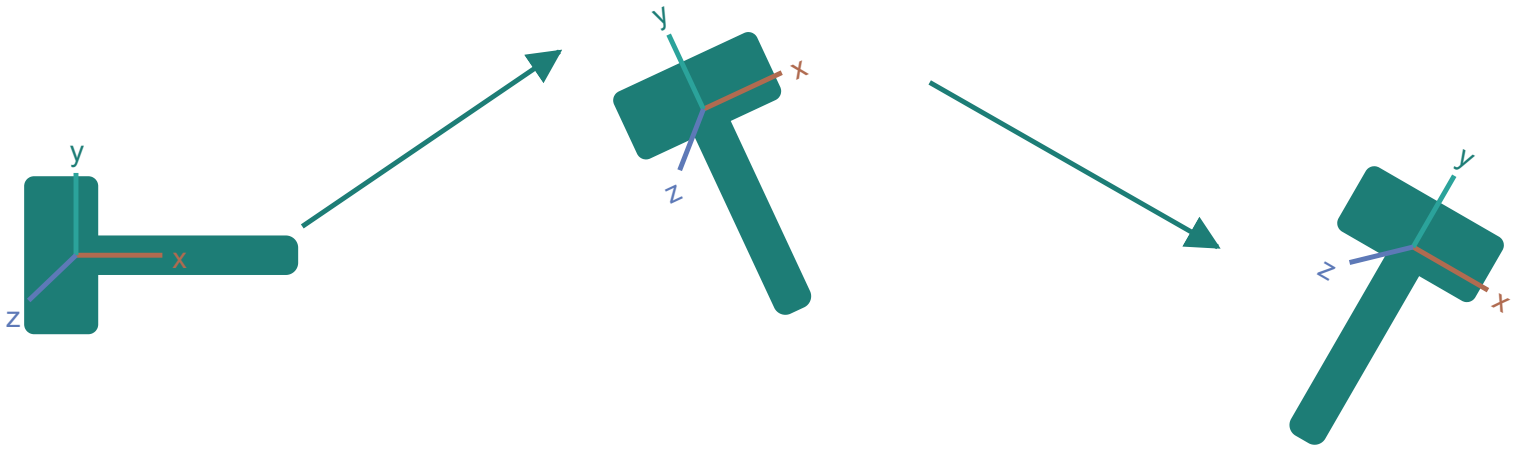
Robot · infeasible grasp

> A demonstrated motion does not automatically become a feasible, stable robot grasp.

02 / METHOD

A 6D pose specifies position and orientation

A task becomes an ordered set of rigid-object configurations



g¹ Lift

g² Reorient

g³ Swing

3D
position

3D
orientation

$$g^k = (R^k, p^k) \in SE(3), \qquad \mathcal{G} = \{g^1, \ldots, g^K\}$$

02 / METHOD

Tool use as a sequence of pose-reaching problems

TASK SPECIFICATION

Human tool motion

6D pose waypoints

→

Reusable goal-reaching RL policy

→

ROBOT EXECUTION

Arm + hand actions

Specify where the tool goes

3D position + 3D orientation

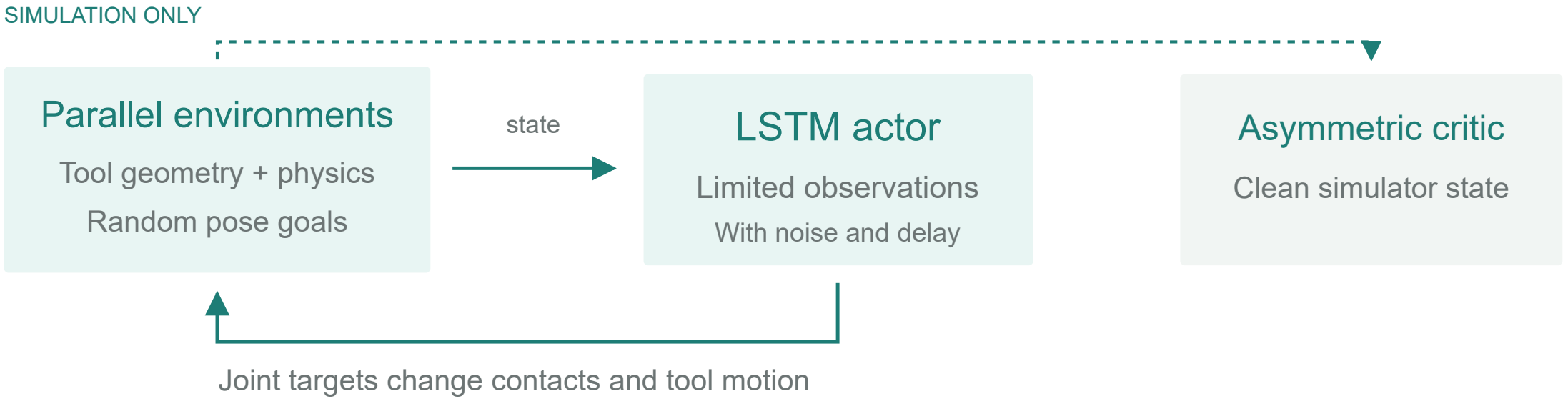
Learn how this robot moves it

Contacts and regrasping emerge from RL

> Object goals specify the motion. RL discovers robot-specific grasps and in-hand actions.

02 / SIMULATION

Training pipeline: learn pose reaching through interaction



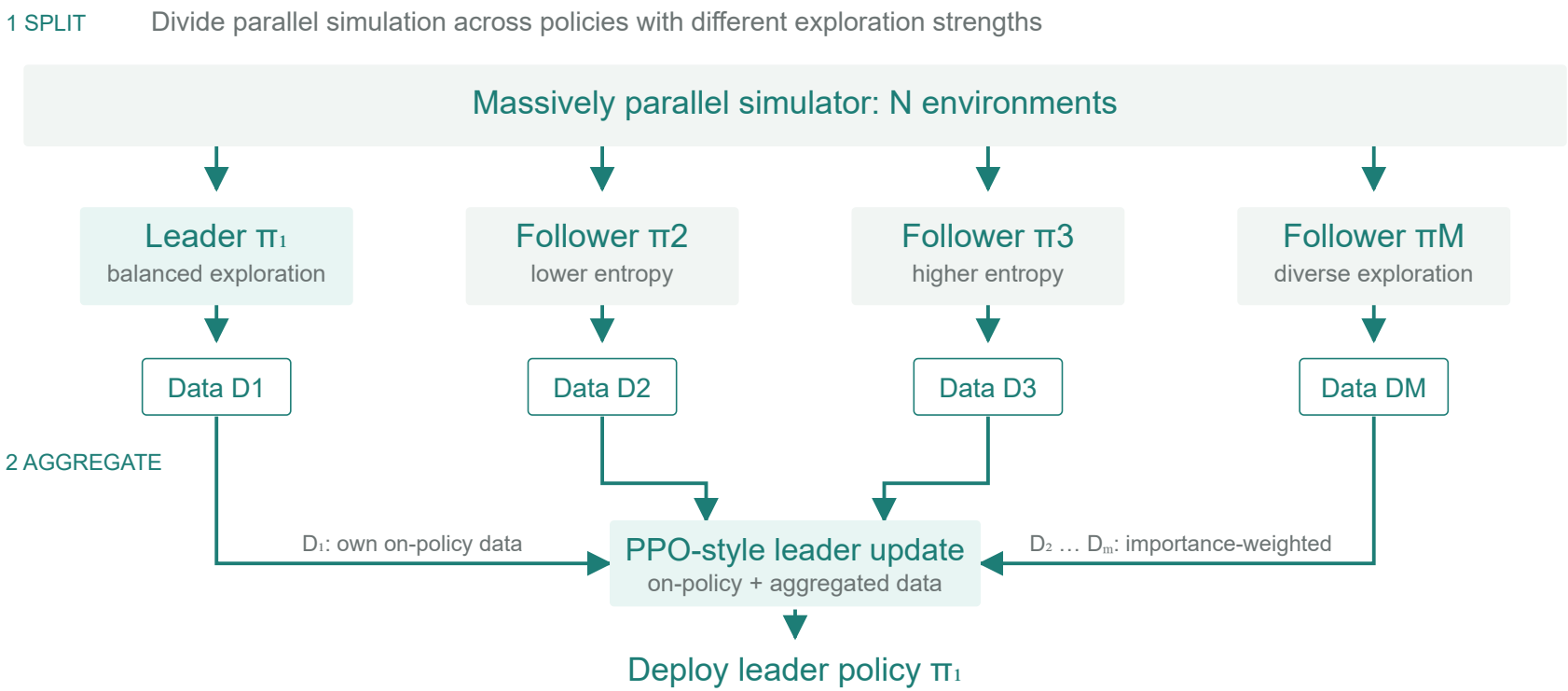
SAPG updates the policy using simulated experience

Critic supports training only

02 / SIMULATION

SAPG splits exploration and aggregates experience

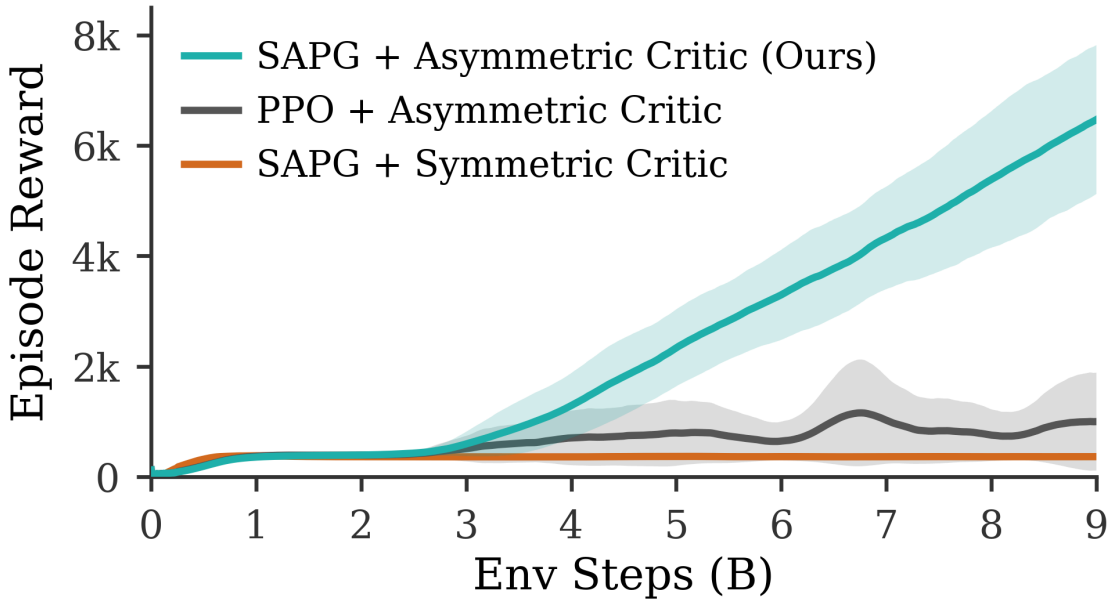
One PPO policy can collect similar experience across thousands of parallel environments.



SAPG retains PPO-style actor–critic updates while changing how experience is explored and reused.

02 / SIMULATION

SAPG learns more effectively than PPO in this setting



Paper Fig. 8 · Training reward, mean over five seeds

Controlled comparison

- Both runs retain the asymmetric critic.
- Replacing SAPG with PPO changes the exploration and data aggregation scheme.

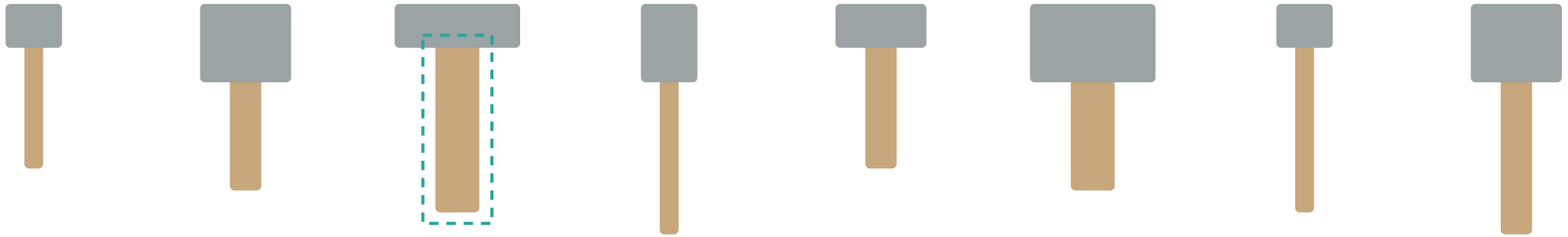
Observed result

- SAPG continues improving across 9B environment steps.
- PPO reaches a much lower plateau under the matched setup.

02 / METHOD

Procedural tools cover shape and mass variation

One structural prior, many physical instances



Geometry

Shape + size

Mass distribution

Handle / head densities

Shared policy inputs

Pose + grasp-region box

The commonality is a geometric interface: tool pose and a graspable region.

02 / SIMULATION

RL training signal: how each reward term is constructed

$$r = r_{\text{smooth}} + r_{\text{grasp}} + \mathbb{I}_{\text{grasped}} r_{\text{goal}}$$

1 · Smooth motion

$$-\lambda_a \|\dot{q}_a\|_1 - \lambda_h \|\dot{q}_h\|_1$$

Penalize current arm and hand joint speeds.

2 · Acquire the grasp

$$r_{\text{approach}} + (1 - \mathbb{I}_{\text{grasped}}) r_{\text{lift}}$$

Approach: best-so-far fingertip distance improvement.

Lift: height progress plus a one-time bonus.

3 · Reach each goal pose

$$\Delta d^* + B_{\text{succ}} \mathbb{I}[d(o_t, g) < \epsilon]$$

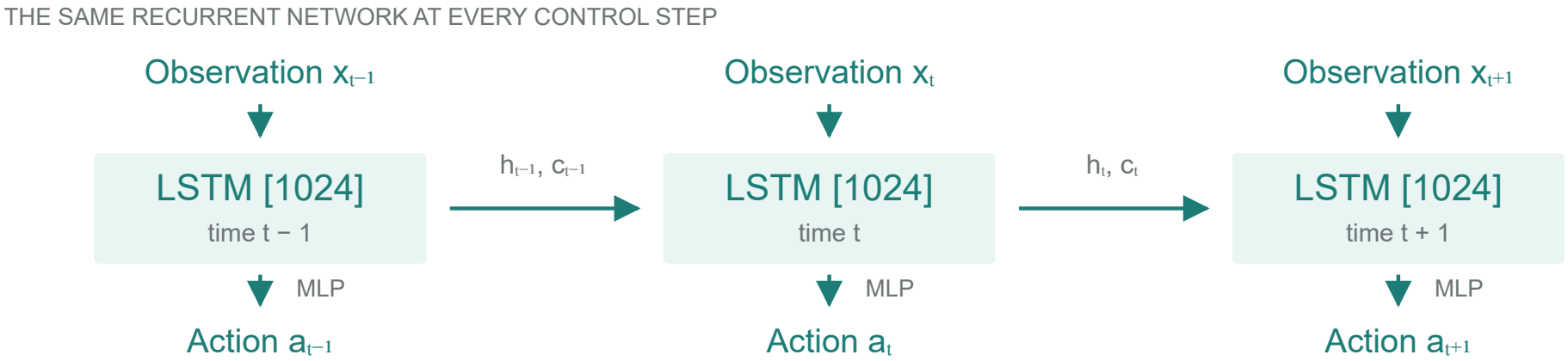
Progress: improve the four-keypoint pose error.

Success: cross ε, receive a bonus, then sample the next goal.

These are environment reward terms. SAPG still optimizes an actor–critic objective from the resulting returns.

02 / LSTM POLICY

LSTM memory carries interaction history across steps



Cell state c_t: internal memory

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Forget gate f_t retains old information. Input gate i_t writes candidate memory.

Hidden state h_t: current representation

$$h_t = o_t \odot \tanh(c_t)$$

Output gate o_t exposes information to the action MLP and the next time step.

02 / LSTM POLICY

The action head controls the arm and hand differently

$$a_t = \left[a_t^{\text{arm}}, a_t^{\text{hand}}\right] \in [-1, 1]^{29}$$

7 arm joints: incremental targets

$$\hat{q}_t^{\text{arm}} = q_{t-1}^{\text{target,arm}} + 0.025 a_t^{\text{arm}}$$

Each action adjusts the previous target.

22 hand joints: absolute targets

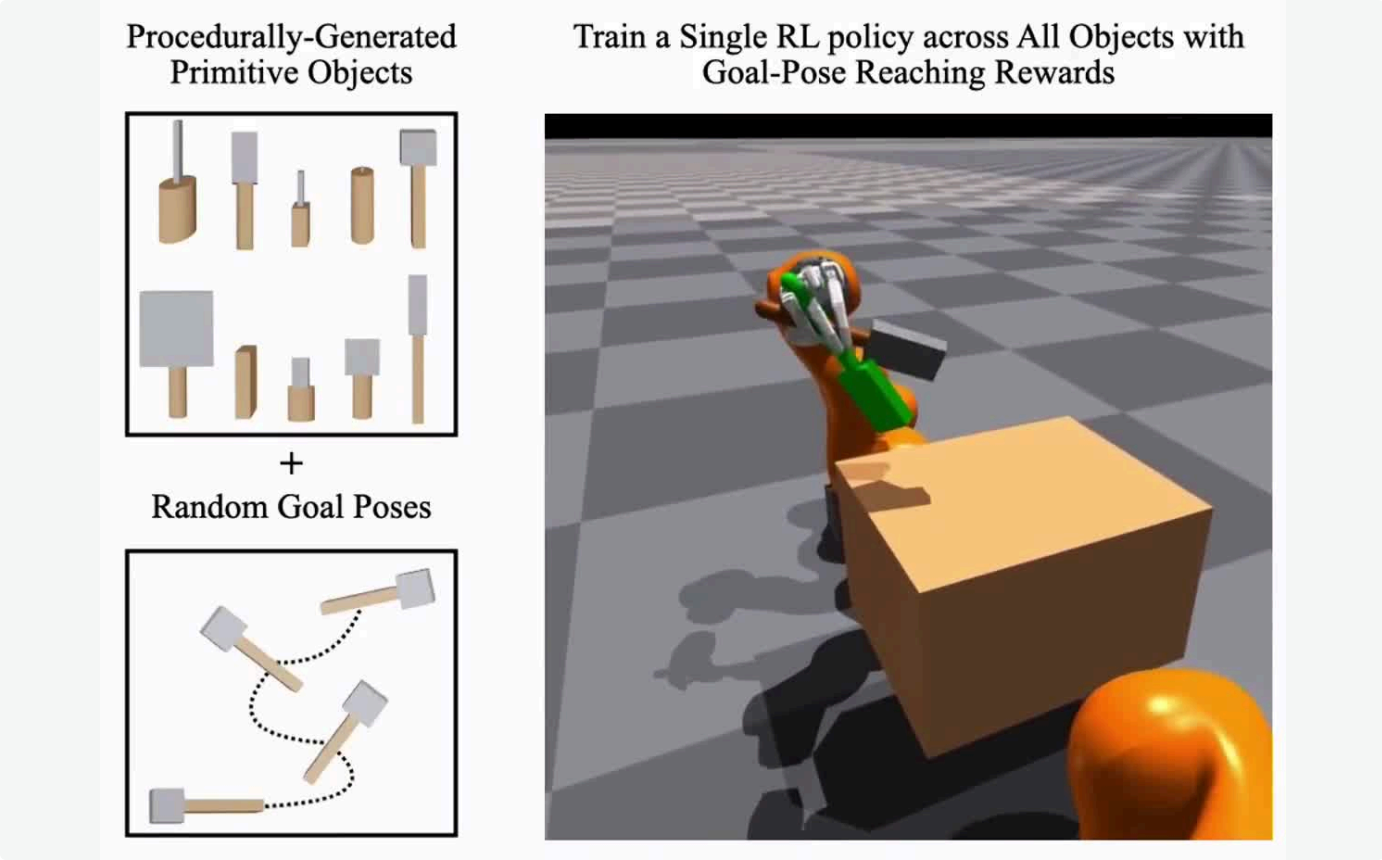
$$\hat{q}_t^{\text{hand}} = q_{\min} + \frac{a_t^{\text{hand}} + 1}{2} \odot (q_{\max} - q_{\min})$$

Each action maps into the joint's allowed range.

> Joint limits and EMA smoothing (α = 0.1) produce the final targets at 60 Hz.

02 / SIMULATION

Simulation: reaching diverse tool poses



WHAT TO WATCH

The **green mesh** indicates the current target pose.

The policy learns to acquire and reorient procedurally generated tools.

There is no human task trajectory in this training loop.

03 / SIM-TO-REAL

Simulation exposes the state. Reality requires estimation.

Inside the simulator

- Exact tool pose and geometry are available.
- Random goal poses define the task.
- Noise and delays train the actor on imperfect inputs.

On the real robot

- The tool pose is hidden behind pixels and occlusion.
- A human demonstration supplies a meaningful trajectory.
- Perception must recover the same policy interface.

> Keep the motor policy fixed. Replace simulator-derived inputs with measured and estimated inputs.

03 / DEPLOYMENT

Real deployment supplies the same policy interface

SETUP FOR A NEW TOOL / DEMONSTRATION

Human RGB-D video
One demonstrated motion

→

Prompted SAM 2 + SAM 3D
Metric mesh + grasp box

→

FoundationPose on the video
Sequence of 6D goal poses

Current goal ↓

ONLINE EXECUTION

Live RGB-D + object mesh
FoundationPose: current tool pose

→

Frozen LSTM policy
Proprioception + grasp box

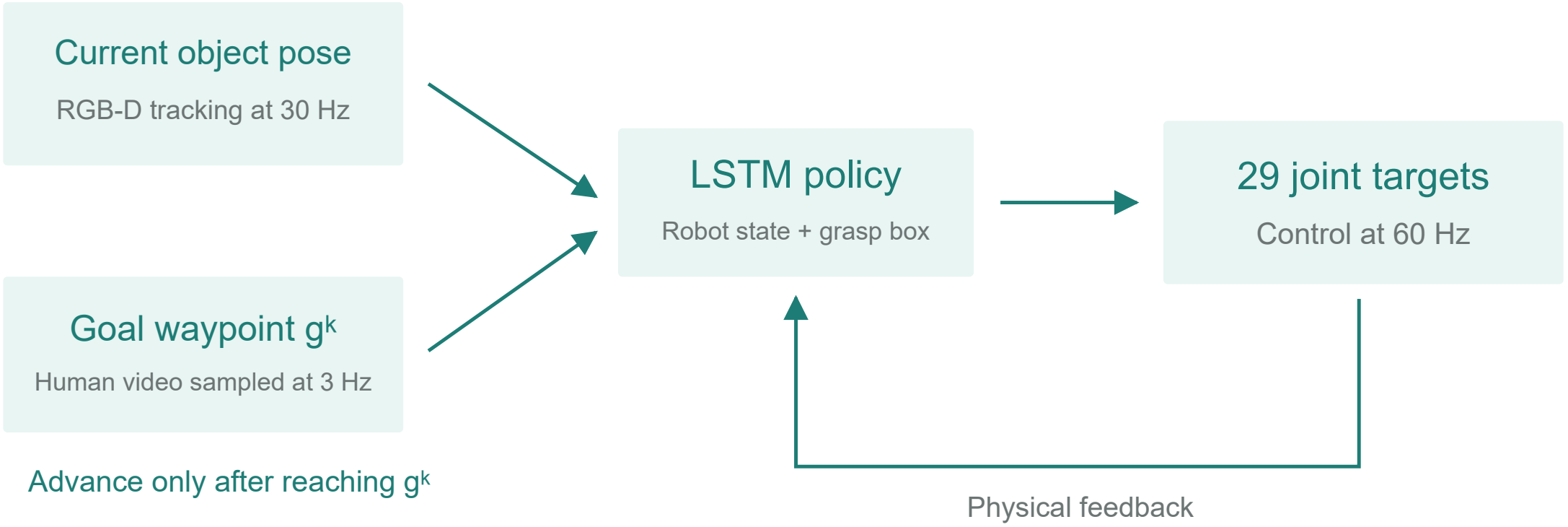
→

Arm + hand
60 Hz control

Physical feedback supplies the next observation

03 / DEPLOYMENT

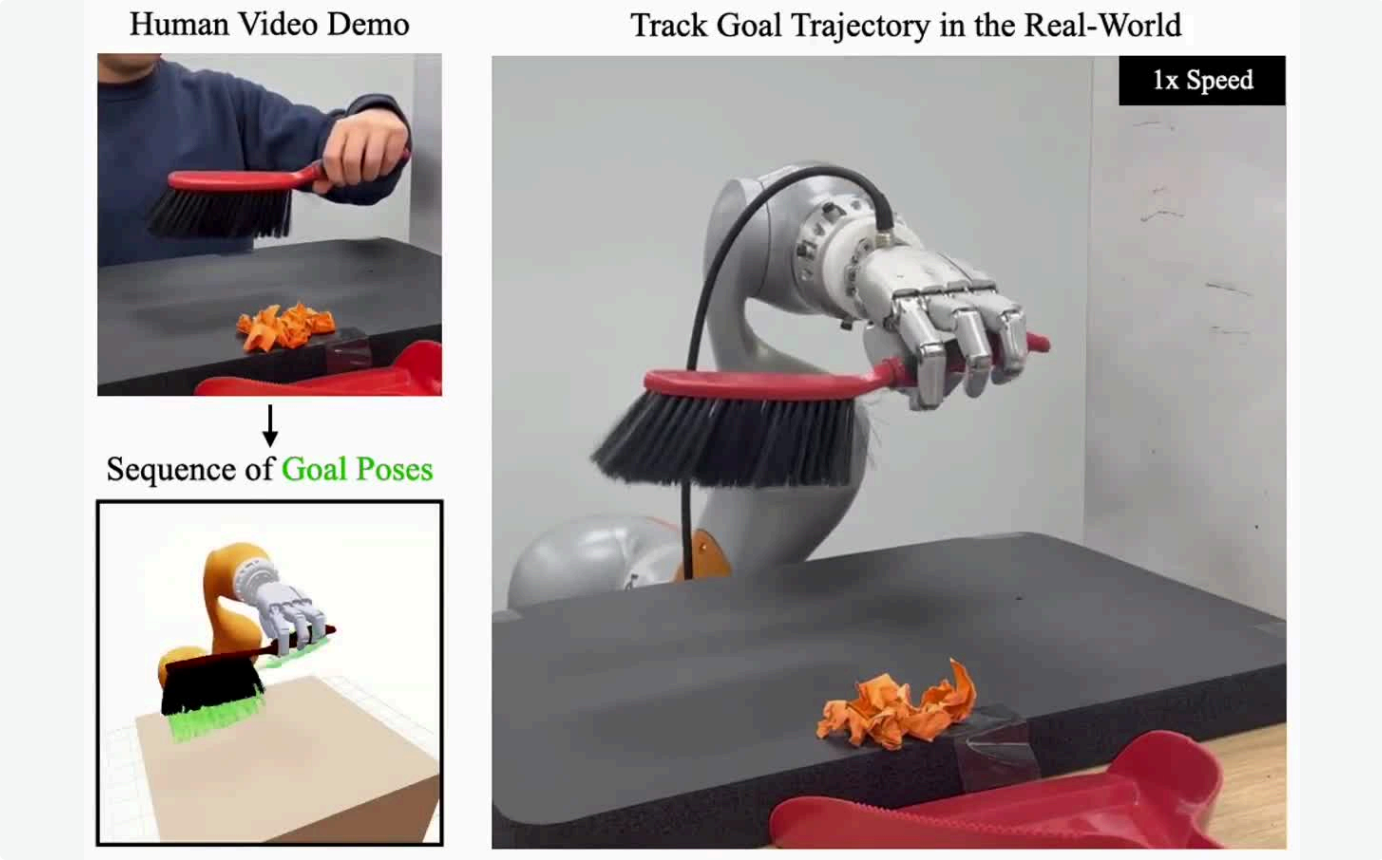
The robot advances goals only after reaching them



3 Hz waypoints · **event-driven** goal switch · **60 Hz** control

03 / DEPLOYMENT

The closed loop follows unseen tool trajectories



WHAT TO WATCH

The human demonstration supplies the **tool-motion goals**.

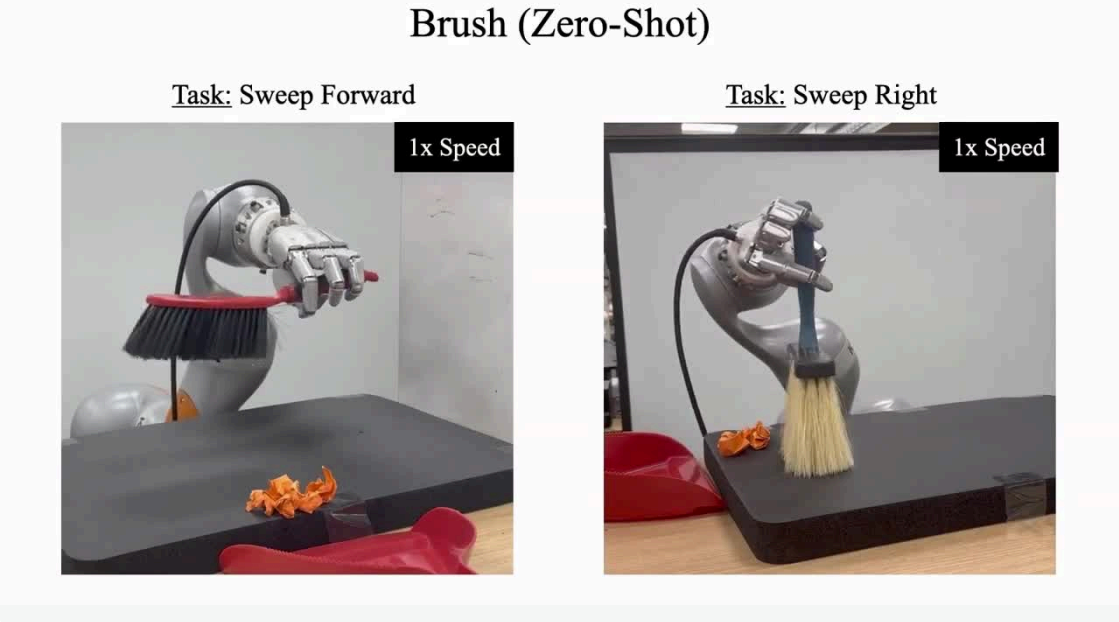
The fixed policy grasps, reorients and moves the real brush.

The goal interface specifies tool motion rather than robot finger trajectories.

04 / REAL-WORLD BEHAVIOR

Brush and hammer: reorientation before interaction

Brush



Reorient the handle, then sweep along the surface.

Hammer

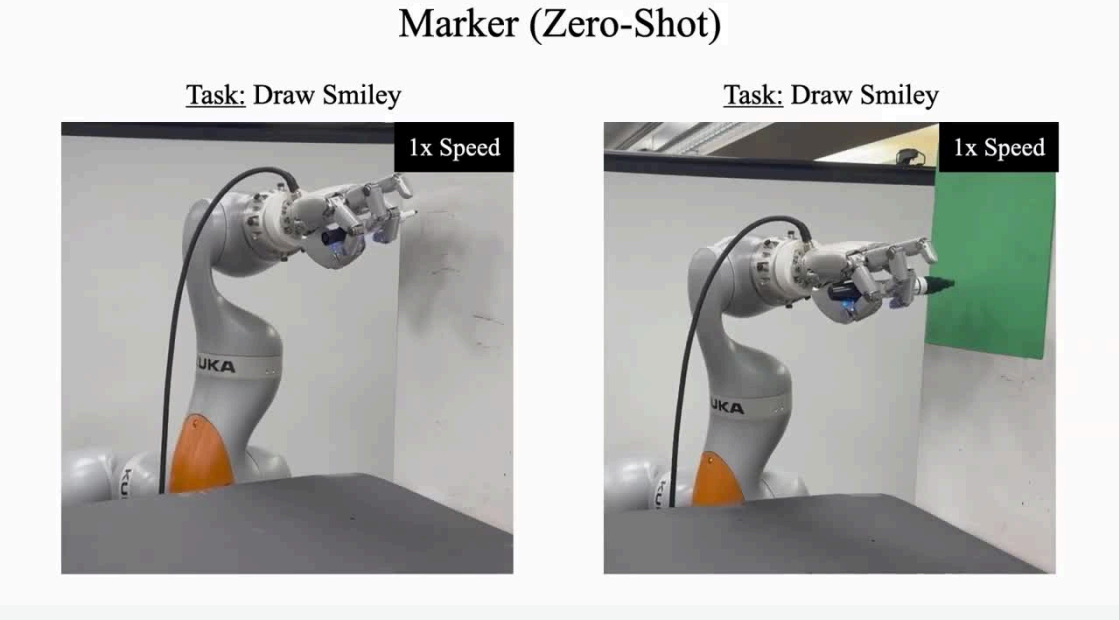


Change the grasp–tool orientation before the swing.

04 / REAL-WORLD BEHAVIOR

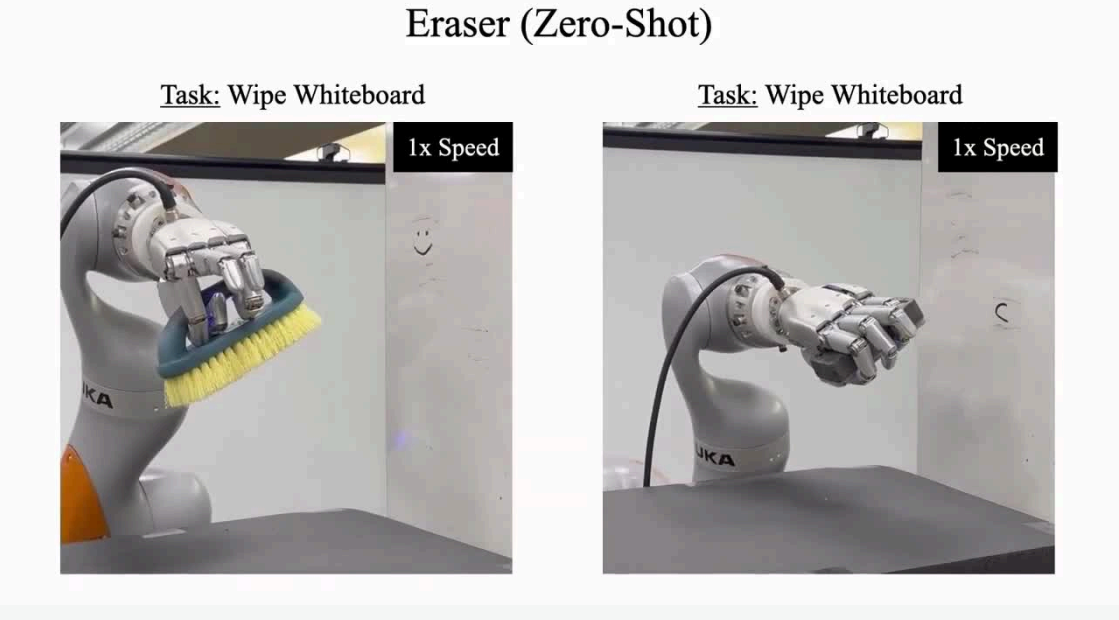
Marker and eraser: tool motion along a surface

Marker



Acquire a thin tool and move its tip across the board.

Eraser

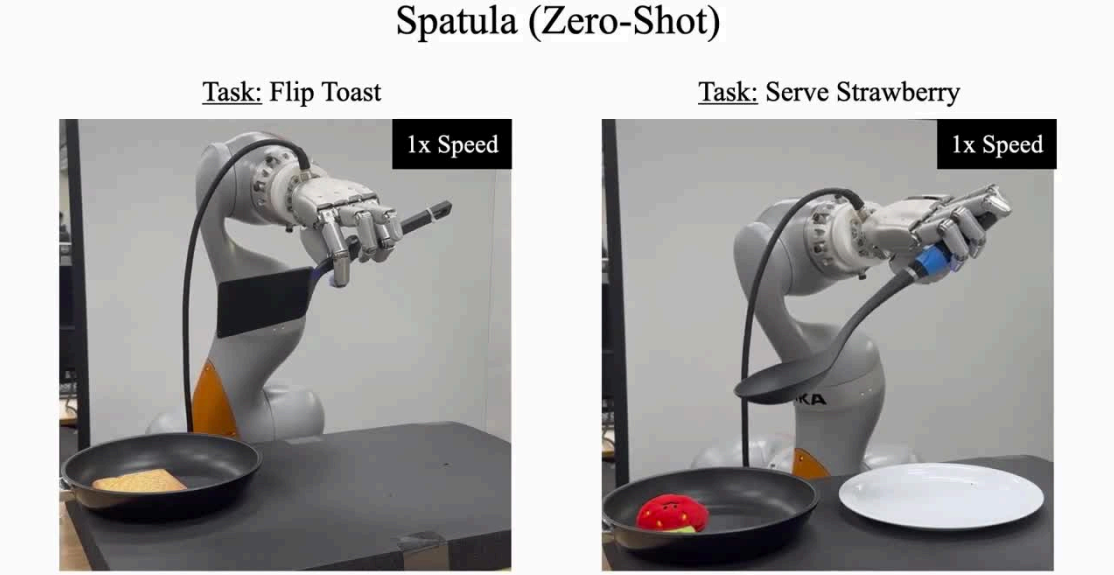


Maintain a grasp while following a wiping path.

04 / REAL-WORLD BEHAVIOR

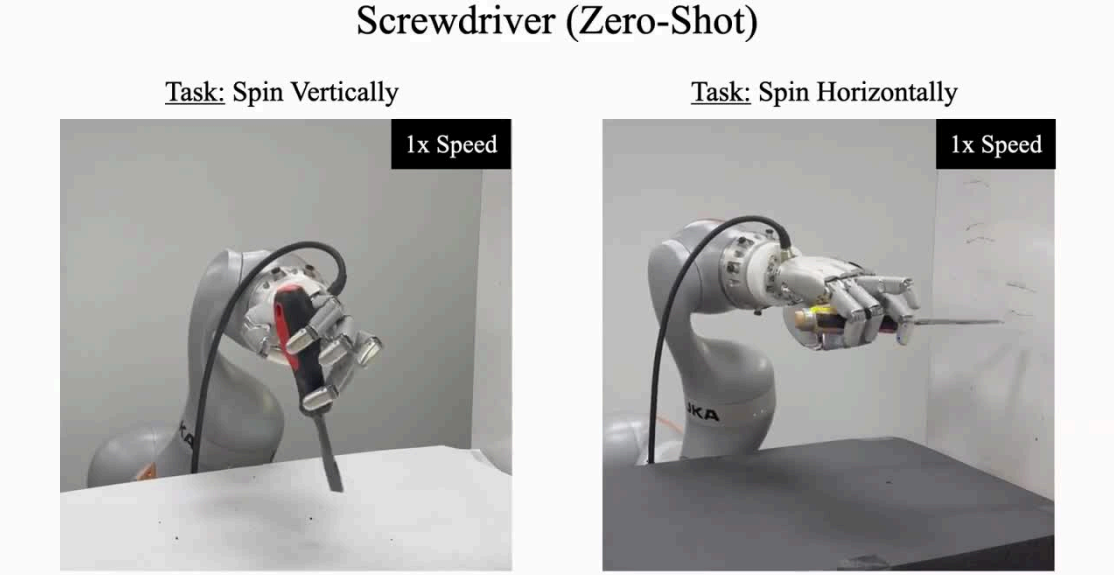
Spatula and screwdriver: larger orientation changes

Spatula



Serving and flipping demand changing tool orientation.

Screwdriver

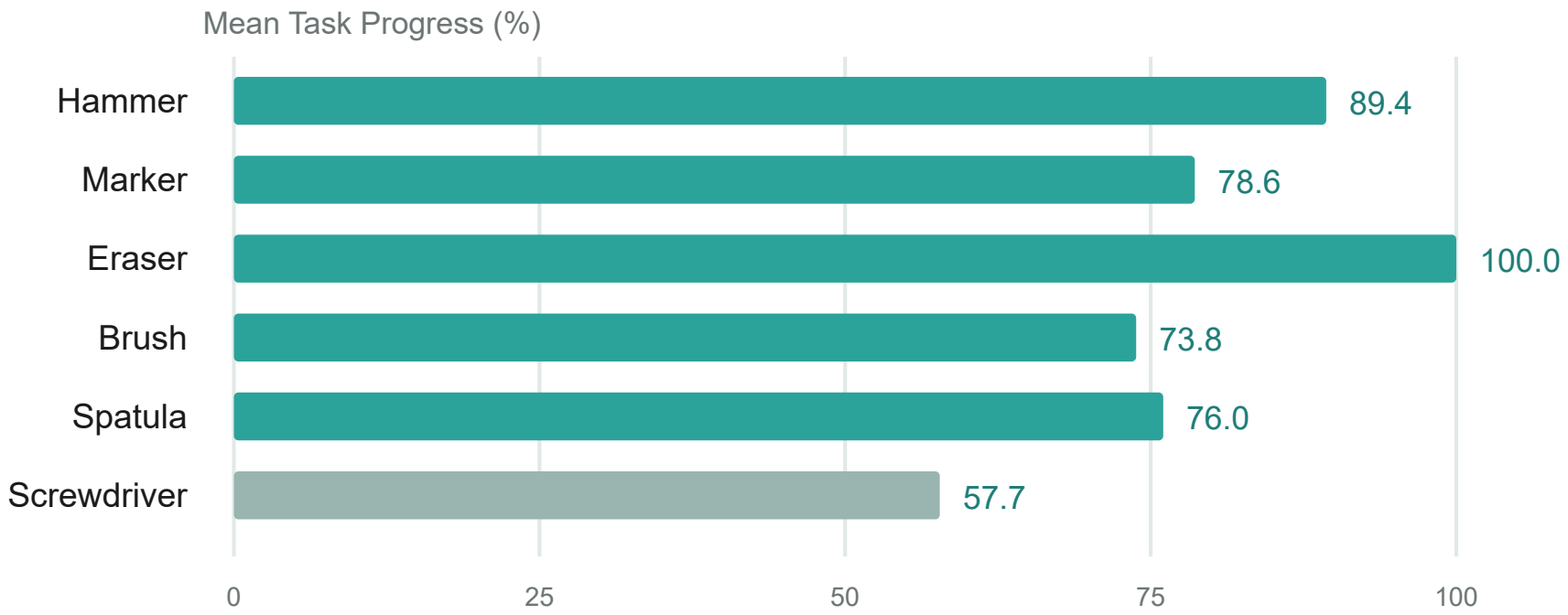


Reorient and spin the tool in free space.

04 / EVIDENCE

Real-world transfer across six tool categories

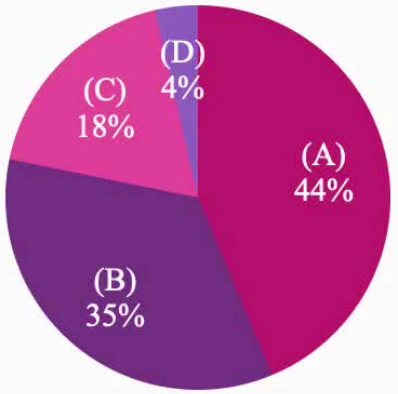
120 rollouts · Task Progress = pose waypoints reached



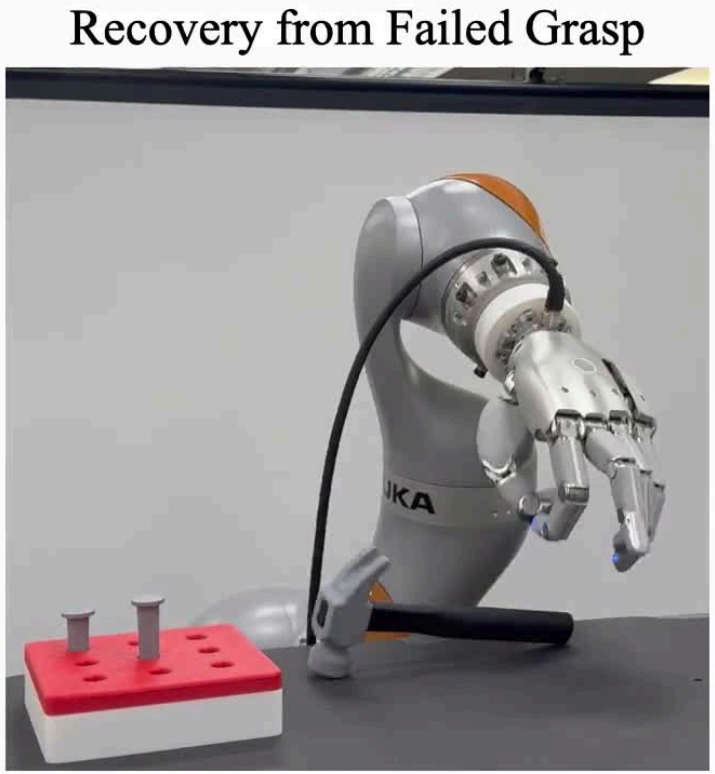
79.3% mean trajectory progress across six categories.

04 / SCOPE

Recovery illustrates feedback, with clear limits



(A) Pose Tracking Failure
(B) Object Drop
(C) Incomplete Rotation
(D) Failed Grasp



WHAT TO WATCH

The policy can attempt to **regrasp** after an error.

Recovery depends on the tool staying **reachable and trackable**.

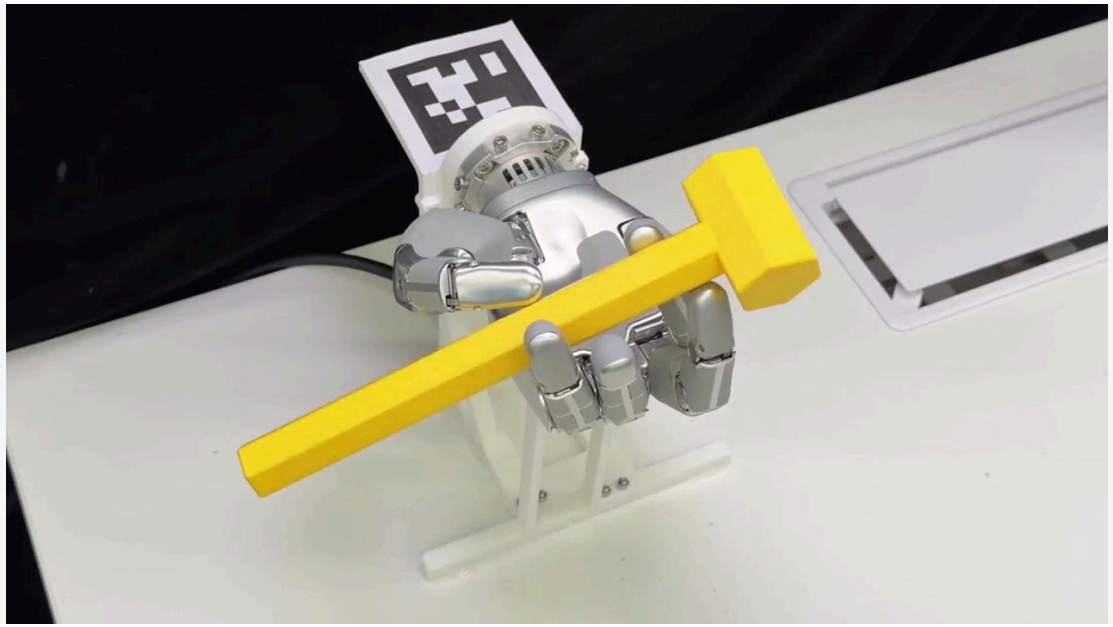
Rigid pose goals leave force objectives, clutter and task replanning unresolved.

05 / RELATED WORK

Object-centric manipulation beyond SimToolReal

POISE · September 2026

Learning In-Hand Object Reaching to General 6D Poses

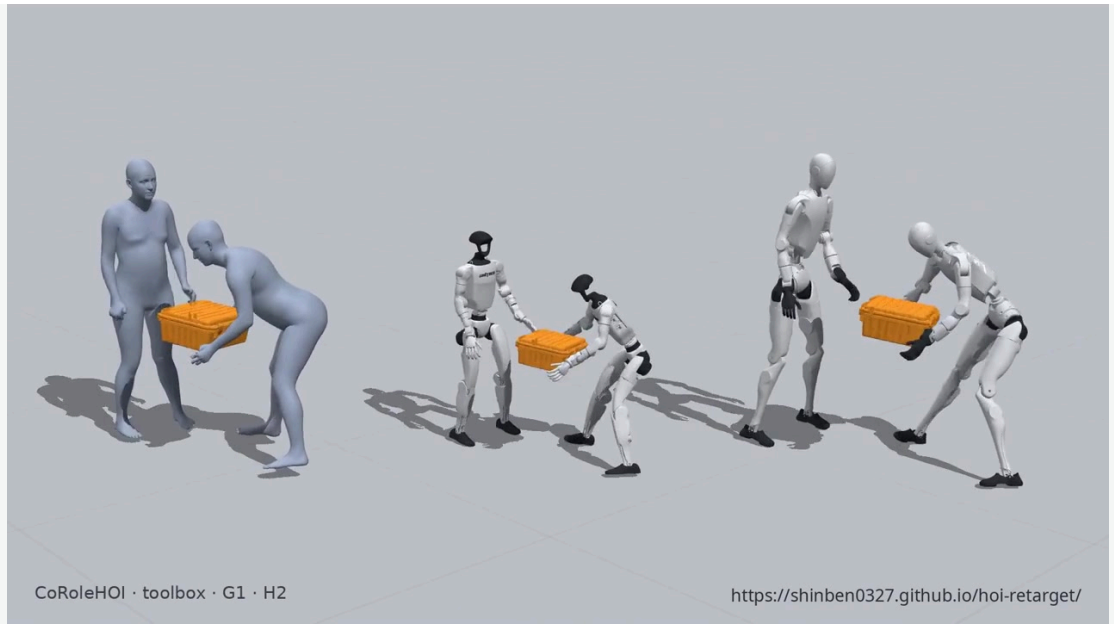


Control the object pose. Fingers move a grasped object toward palm-relative 6D targets.

Real robot · Paper · Project / video

HOI-Retarget · ETH Zürich · September 2026

HOI-Retarget: Contact-Centric Retargeting for Human-Object Interaction



Preserve object contacts. Object-frame targets retain contact location and timing across embodiments.

Retargeting visualization · Paper · Project / video

05 / DISCUSSION

Structured simulation as a route to reusable robot skills

PRESENTER REFLECTION

01 · PRIMARY TAKEAWAY

Beyond scaling end-to-end action models, structured simulation may offer another path toward skill acquisition in the wild.

RL learns reusable motor competence. Perception, demonstrations or a higher-level model specify what the tool should do.

02 Compact interfaces beyond tool use

Search for compact task interfaces—adding contact, force or articulation when pose is insufficient.

03 Lightweight VLMs for semantic setup

Use a VLM for tool selection and subgoals; preserve geometric tracking for metric feedback.

BACKUP

References, videos and source material

Main paper

Kedia, Lum, Bohg and Liu. *SimToolReal: An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation.*
arxiv.org/abs/2602.16863v2 · RSS 2026 proceedings

Project and code

simtoolreal.github.io · github.com/tylerlum/simtoolreal

Closely related work

Chen et al. *Object-Centric Dexterous Manipulation from Human Motion Data.* CoRL 2024.
Lum et al. *Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL Using One Human Demonstration.* 2025.
Singla et al. *Split and Aggregate Policy Gradients.* ICML 2024.
Petrenko et al. *DexPBT.* 2023.

Design reference

Mizoreww/steerable-vla-talk